Software Requirements Specification

for

Social Media Management System

Version 1.0 approved

Prepared by Mahima Kumari

Silicon University

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Revision History

Name	Date	Reason For Changes	Version

1. Introduction

1.1 Purpose

This document specifies the software requirements for the Social Media Management System, Revision 1.0. The scope covers the core system architecture, including user authentication, the campaign management module for content creators, the view verification and tracking engine, and the internal wallet and ledger system for micro-payouts.

1.2 Document Conventions

This SRS follows standard IEEE documentation conventions. Bold text is used for defining key system modules and database entities. Requirement priority levels (High, Medium, Low) are explicitly stated for each functional requirement rather than assumed to be inherited from higher-level features.

1.3 Intended Audience and Reading Suggestions

This document is intended for software developers, project managers, database architects, QA testers, and system administrators. Section 1 provides the product overview, Section 2 details system constraints and user classes, and Section 3 outlines specific functional and non-functional requirements. Developers should focus on Section 3, while project managers and testers should review sections 1 and 2 first.

1.4 Product Scope

The software is a two-sided marketplace platform designed to connect social media content creators with users who earn micro-monetary rewards by watching and engaging with content. The primary objectives are to facilitate creator campaign funding, automate view verification, prevent bot fraud, and manage secure micro-transactions, directly supporting the business goal of a sustainable, fraud-resistant engagement model.

1.5 References

- *IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications.*
- *Project Vision and Scope Document, Version 1.0, internal development repository.*
- *OAuth 2.0 Authorization Framework, RFC 6749.*

Name – Mahima Kumari
SIC – 24BCSF87
Roll No. – 24
Date – 3rd Sept 2026

2. Overall Description

The Social Media Management System performs several major functions to connect content creators and reward-seeking viewers. At a high level, the system must allow creators to deposit funds, create and manage social media promotion campaigns, and monitor view analytics. It must enable viewers to browse curated content, watch videos for required durations, and track their accumulated micro-earnings. Additionally, the system must execute automated view verification, prevent bot fraud and metric inflation, and process secure internal wallet transactions and payouts.

2.1 User Classes and Characteristics

- ***Content Creators (Advertisers):*** *Users who fund campaigns to increase visibility on their social media accounts. They possess moderate technical literacy, use the platform frequently to manage budgets, and hold high security privileges regarding financial deposits and campaign controls.*

- ***Viewers (Earners):*** *Users who watch content to earn small monetary amounts. They represent a broad demographic with varying technical expertise, use the platform primarily for micro-earnings, and have standard user privileges restricted to viewing, verification completion, and wallet withdrawals.*

- ***Platform Administrators:*** *Internal staff responsible for monitoring system health, reviewing fraud flags, managing user disputes, and overseeing platform-wide transactions. They require high technical expertise and full administrative security privileges.*

2.2 Operating Environment

The software will operate in a cloud-hosted environment supporting scalable containerized microservices. The backend will run on Linux-based cloud servers (e.g., AWS or Google Cloud Platform), utilizing a relational database for secure financial ledgers and user accounts. The client-facing applications will include a responsive web application compatible with modern desktop and mobile browsers (Chrome, Safari, Firefox, Edge) and potentially companion mobile applications built for Android and iOS.

2.3 User Documentation

- *Creator User Manual: Comprehensive guide detailing how to set up accounts, deposit funds, create campaigns, and track analytics.*
- ***Viewer Help Guide & FAQ:*** *Instructions on how to browse content, fulfill watch-time verification requirements, and request wallet withdrawals.*
- ***In-App Tooltips and Onboarding Walkthroughs:*** *Interactive guides embedded within the user interface to assist new creators and viewers during their first session.*
- ***API and Integration Documentation:*** *Technical documentation for administrative staff and third-party API integrations (such as payment gateways and social media platforms).*

3. External Interface Requirements

3.1 User Interfaces

The platform requires distinct user interfaces for Content Creators, Viewers, and Administrators. All screens must follow responsive design standards to support desktop, tablet, and mobile browsers. Common elements include persistent navigation bars, standardized error message banners, and wallet balance indicators.

- ***Creator Dashboard:*** *Displays campaign metrics, budget allocations, deposit portals, and performance charts.*
- ***Viewer Feed Interface:*** *Features a content playback window, a real-time watch-time countdown timer, and reward tracking indicators.*
- ***Admin Panel:*** *Provides system-wide monitoring, user management tables, and fraud flag review queues. Detailed screen mockups and style guides are maintained in a separate UI specification document.*

4. System Features

The functional requirements for the Watch Social Media Management System Platform are organized by major system features, capturing the core services provided to content creators, viewers, and administrators.

4.1 Campaign Management Module

4.1.1 Description and Priority

Allows content creators to create, fund, pause, and track promotional campaigns for their social media links.

- *Priority: High*
- *Benefit: 9 | Penalty: 8 | Cost: 7 | Risk: 6*

4.1.2 Stimulus/Response Sequences

- *Stimulus: Creator inputs campaign details (target URL, view counts, budget) and submits funds.*
- *Response: System validates the account balance, deducts the funds into an escrow ledger, and publishes the campaign to the active viewer queue.*
- *Stimulus: Creator clicks "Pause Campaign."*
- *Response: System immediately removes the campaign from active viewer feeds and updates status.*

4.1.3 Functional Requirements

- *REQ-1: The system shall allow creators to input social media links from supported platforms.*

- *REQ-2: The system shall calculate and display the total campaign cost based on desired view counts and platform rates.*
- *REQ-3: The system shall automatically transition a campaign to "Completed" status once the target view count is reached.*
- *REQ-4: The system shall handle insufficient creator balances by displaying an error message and blocking campaign activation.*

4.2 Watch-Time Verification and Reward Engine

4.2.1 Description and Priority

Tracks viewer engagement and verifies that content is watched for the required duration before crediting micro-payouts.

- *Priority: High*
- *Benefit: 9 | Penalty: 9 | Cost: 8 | Risk: 8*

4.2.2 Stimulus/Response Sequences

- *Stimulus: Viewer clicks on a video card and initiates playback.*
- *Response: System starts a background timer and monitors focus state to ensure the tab remains active.*
- *Stimulus: Viewer successfully completes the required watch duration without interruption.*
- *Response: System credits the micro-amount to the viewer's internal wallet and loads the next content item.*

4.2.3 Functional Requirements

- *REQ-5: The system shall pause the watch timer automatically if the user switches browser tabs or minimizes the window.*
- *REQ-6: The system shall verify unique views per user to prevent duplicate farming of the same content piece.*
- *REQ-7: The system shall update the viewer's internal wallet ledger in real time upon successful verification.*
- *REQ-8: The system shall trigger automated fraud detection scripts if abnormal view velocity or automated click patterns are detected.*

5. Other Nonfunctional Requirements

5.1 Performance Requirements

The platform must support high-concurrency environments to handle thousands of simultaneous viewers and active campaign tracking. The system must respond to user interface actions, such as loading video feeds and initiating timers, in under 2 seconds. The watch-time verification engine

must process and log heartbeats and balance updates with a latency of no more than 500 milliseconds. Database queries for financial ledgers and campaign queues must execute under peak loads without dropping transactions.

5.2 Safety Requirements

The system must incorporate safeguards to prevent financial loss or data corruption resulting from unexpected system crashes, network interruptions, or payment gateway failures. All funds in escrow must be safely isolated from operational accounts to protect creator deposits during system downtimes. The platform must prevent negative account balances and ensure automatic transaction rollbacks if a reward payout or deposit fails mid-process.

5.3 Security Requirements

All user authentication must enforce strong password policies and support secure token-based sessions using JSON Web Tokens (JWT) with HTTPS encryption enforced across all endpoints. Financial data, user credentials, and personal information must be encrypted at rest and in transit. The system must restrict administrative access through Role-Based Access Control (RBAC) and comply with data privacy standards regarding user tracking and telemetry storage.

Appendix A: Glossary

- ***SRS:*** *Software Requirements Specification.*
- ***Escrow:*** *A temporary holding account where creator funds are safely locked until view verification metrics are successfully met.*
- ***Micro-payout:*** *Small monetary rewards credited to viewer wallets upon successful completion of content watch-time requirements.*
- ***Device Fingerprinting:*** *A technique combining browser attributes, hardware parameters, and network data to uniquely identify user devices and block multi-accounting fraud.*
- ***Heartbeat:*** *A periodic background signal sent from the client viewer to the server to verify active engagement and tab focus.*

Appendix B: Analysis Models

- *Data Flow Diagram (DFD): Illustrates the flow of money from creator deposits into escrow, down to viewer micro-payouts and platform commission deductions.*
- ***Entity-Relationship (ER) Diagram:*** *Maps the structural relationships between Users, Campaigns, Transactions, Wallets, and Verification Logs.*
- ***State-Transition Diagram:*** *Details the lifecycle states of a promotional campaign (Draft, Active, Paused, Completed, Cancelled).*